

Department of Mathematics and Applied Mathematics
MAM2000W: 2LA (Linear Algebra)

Test 1

17 May 2021

Time : 3 hours

Maximum marks: 51

Full marks: 50

- This question paper consists of 4 pages (including this one).
- For Section A, only final answers will be marked. For Section B, full answers must be given.
- Be careful to provide answers that we can read and make sense of at all times. We will pay attention to your presentation as well as the content. Work that is poorly presented will be penalized. If we can't read your handwriting, we will mark your solution incorrect.

Section A

In this section, give your final answer only. Number the answers clearly. Do not show your working.

Question 1. [2 points] The 3×3 matrix A is skew-symmetric and $a_{31} = 3, a_{23} = 7$ and $a_{12} = 4$. Find A .

Question 2. [6 points] Let $\mathcal{U}, \mathcal{E}$ and $\mathcal{I}$ respectively denote the set of all 3×3 matrices that are upper triangular, in row echelon form and invertible. Let

$$A_1 = \begin{pmatrix} 1 & 0 & 2 \\ 0 & -1 & 0 \\ 0 & 0 & 3 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 2 & 3 & 1 \\ 0 & 0 & 4 \\ 0 & 0 & -1 \end{pmatrix},$$
$$A_3 = \begin{pmatrix} 0 & -1 & 0 \\ -1 & 0 & 2 \\ 0 & 2 & 5 \end{pmatrix}, \quad A_4 = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Give an example, if possible, of an A_i ($1 \leq i \leq 4$) such that

- (a) $A_i \in \mathcal{U}$ but $A_i \notin \mathcal{I}$;
- (b) $\{A_i, A_i^T\} \subseteq \mathcal{U}$;
- (c) $A_i = A_i^T$, $A_i \notin \mathcal{U}$ and $A_i \in \mathcal{I}$;
- (d) $A_i \in \mathcal{U}$ but $A_i \notin \mathcal{E}$;
- (e) A_i is the product of two elementary matrices.

Choose from A_1, A_2, A_3, A_4 or “Impossible” for your answers. You may use a matrix more than once, or not at all.

Question 3. [6 points] Consider the following elementary row operations:

e_1 : Add 5 times row 1 to row 3.

e_2 : Multiply row 2 by $-\frac{1}{3}$.

e_3 : Interchange row 2 and row 3.

Let E_i be the 3×3 elementary matrix corresponding to e_i , for $i = 1, 2, 3$, and let A be a 3×3 matrix such that

$$e_3(e_2(e_1(A))) = \begin{pmatrix} 1 & 0 & -5 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (**)$$

(a) Find E_1 , E_2 , E_3 and their inverses.

(b) Find A .

Section B

In this section, give full answers to all the questions. Show all your working.

Question 4. [4 points] Let A be the 3×3 matrix described in Question 3. Use the equation (**) in Question 3 to determine whether A is invertible or not. Explain clearly how you arrived at your answer. If A is invertible, write it as a product of elementary matrices. Do not multiply out the product.

Question 5. [8 points] The augmented matrix of a system of linear equations is

$$\left(\begin{array}{cccc|c} 1 & -6 & 3 & 3 & 9 \\ 0 & 2 & 0 & -1 & 3 \\ 0 & 4 & 0 & -2 & 6 \\ 0 & 0 & -1 & 6 & 1 \\ 0 & 0 & k & k^2 & 6 \end{array} \right), \text{ where } k \text{ is a constant.}$$

Find a row-echelon form for this augmented matrix. Then use this to answer the following questions, saying how you arrived at your answer.

- (a) For which values of k , if any, will the system have no solution?
- (b) Give the values of k , if any, for which the system will have infinitely many solutions. If there are such values of k , give the solutions in vector form.
- (c) For which values of k , if any, will the system have a unique solution?

Question 6. [6 points] Consider the statement:

For every square matrix A , if $A^T A = I$, then A is non-singular. (*)

- (a) Write down the converse of the statement (*).
- (b) Write down the contrapositive of this statement (*).
- (c) Prove or disprove the statement (*).
- (d) Prove or disprove the converse of the statement (*).

When you are proving that a statement is true, you may use any result proved in lectures, but say which result you are using.

Question 7. [6 points] Let $\begin{pmatrix} 2 & 0 & 0 \\ -2 & -4 & 2 \\ 2 & -1 & -1 \end{pmatrix}$.

- (a) Find all the eigenvalues of A .
- (b) Find an eigenvector of A corresponding to the smallest eigenvalue of A .

Question 8. [13 points] Carefully read the following proof.

Proof: Let A and B be $n \times n$ matrices. We consider two mutually exclusive cases.

Case 1: B is singular. (1)

Then AB is singular as well. (2)

It follows that $\det(AB) = 0 = \det(A) \times 0 = \det(A) \det(B)$. (3)

Case 2: B is non-singular. (4)

Then there are elementary matrices $E_1, E_2, \dots, E_k$ such that $B = E_1 E_2 \dots E_k$ (5)

We know that for any $n \times n$ elementary matrix E and any $n \times n$ matrix C ,

$\det(EC) = \det(E) \det(C)$. (6)

We can use this to prove that for any $n \times n$ elementary matrix E and any $n \times n$ matrix C ,

$\det(CE) = \det(C) \det(E)$. (7)

Therefore,

$$\det(AB) = \det(AE_1 E_2 \dots E_k) \quad (8)$$

$$= \det(AE_1 E_2 \dots E_{k-1}) \det(E_k) \quad (9)$$

$\vdots$

$$= \det(A) \det(E_1) \dots \det(E_k) \quad (10)$$

$$= \det(A) \det(B) \quad (11)$$

Now answer the following questions:

- (a) State the result proved here.

- (b) How does line (2) follow from line (1)?
- (c) How does line (3) follow from the previous lines?
- (d) Explain how line (5) follows from line (4).
- (e) Which previous line do you use to write down line (8)?
- (f) Explain how line (9) follows from line (8).
- (g) Let A be the matrix in Question 3. Use only the equation (**) to find $\det(A)$; do not use your answer for A in Question 3. Show how you obtained your answer.
- (h) Prove the result in line (7) using the result in line (6).